

Factory Safety Standard Operating Procedure

Synthetic reference document - AI Factory Intelligence Command Center. For demonstration only; not a real operating procedure.

1.0 Purpose and Scope

This Standard Operating Procedure defines the mandatory safety steps for operators and maintenance technicians working on CNC milling and turning cells classified as Type L (low-grade), Type M (medium-grade) and Type H (high-grade). It applies to all planned maintenance, unplanned intervention, and inspection activities.

No predictive alert from the Command Center authorises autonomous action. Every intervention requires a named human approver as described in section 6.0.

2.0 Lockout / Tagout (LOTO)

2.1 Before any physical work, isolate the machine at the main disconnect, apply a personal lock, and verify zero energy state at the spindle drive and hydraulic accumulator.

2.2 Rotating spindles must come to a complete mechanical stop. Do not rely on the controller e-stop as an energy isolation device.

2.3 Tag must record technician name, work order number, and expected restoration time.

3.0 Response to Overstrain (OSF) Conditions

3.1 An overstrain condition is indicated when the product of tool wear and torque exceeds the Type-specific threshold (Type L approx 11,000 min*Nm, Type M approx 12,000, Type H approx 13,000).

3.2 On an OSF alert, immediately reduce feed rate and bring the axis load below 50 percent. Do not continue the current load-bearing pass.

3.3 Inspect the tool for chipping and the workpiece for deformation before restarting. Replace the tool if wear is above 200 minutes.

4.0 Response to Heat Dissipation (HDF) Conditions

4.1 HDF risk rises when the air-to-process temperature difference falls below 8.6 K while rotational speed is under 1380 rpm.

4.2 Confirm coolant flow and check the heat exchanger inlet filter. Clear any chip blockage around the spindle housing.

4.3 If the temperature difference cannot be restored within 10 minutes, stop the machine and raise a maintenance work order at HIGH urgency.

5.0 Response to Power (PWF) and Tool Wear (TWF) Conditions

5.1 Power failure risk (PWF) occurs when the mechanical power (torque times angular velocity) leaves the 3.5 - 9 kW band. Check drive tuning and supply voltage; a sustained low-power reading often precedes a stall.

5.2 Tool wear failure (TWF) is expected between 200 and 240 minutes of cumulative tool wear. Schedule a tool change at 200 minutes; never run a tool past 240 minutes.

5.3 Random failures (RNF) have no measurable precursor. Treat any RNF flag as advisory and corroborate with vibration and acoustic inspection.

6.0 Human Approval and Escalation

6.1 The Command Center issues DECISION SUPPORT only. A maintenance supervisor must APPROVE, REJECT or MODIFY every recommendation and record the reason.

6.2 HIGH risk recommendations require sign-off by the shift engineer in addition to the supervisor.

6.3 If a recommendation is rejected, the machine may only keep running under increased monitoring with the residual risk logged in the incident report.

7.0 Personal Protective Equipment

Safety glasses and cut-resistant gloves are mandatory in all machining cells. Hearing protection is required when rotational speed exceeds 2000 rpm. Gloves must not be worn while a spindle is rotating.